Vivek R Gowda

GitHub: github.com/vivek-rg | LinkedIn: linkedin.com/in/vivek-rg | vivekrg2005@gmail.com | +91 7975716776

Summary

Backend-focused Software Engineer with hands-on experience building scalable applications using **Java, Spring Boot, Python, FastAPI, and AWS**. Skilled in REST APIs, secure authentication, microservices, event-driven systems with **Kafka and Redis**, and relational and NoSQL databases. Experienced with **Docker, Azure, AI/LLM integrations**, and delivering backend platforms from development to deployment.

Education

Dayananda Sagar College of Engineering — Expected September 2027
B.E. in Information Science and Engineering — **CGPA: 9.2 / 10.0**

Technical Skills

Languages: Java, Python, C++, C, SQL, JavaScript
Backend & AI: Spring Boot, Spring Security, Spring Data JPA, REST API Design, FastAPI, SQLAlchemy, Google OR-Tools (MILP), LLM Integration (Groq API)
Databases & Messaging: PostgreSQL, MongoDB, Redis, Apache Kafka
Observability & Tools: OpenTelemetry (OTel), Docker, AWS, Azure, Git, GitHub, Linux, JUnit, Mockito, Postman

Projects

TrueNotes | *Java, Spring Boot, MongoDB, Redis, Apache Kafka, OAuth2, Docker, AWS EC2* *GitHub*

- Built a backend for user authentication and real-time updates, engineering **RESTful APIs** covering full CRUD lifecycle for notes.
- Integrated **Google OAuth2** for secure third-party authentication and stateless session handling.
- Reduced repeated database load with **Redis caching** and enabled asynchronous notifications via **Apache Kafka event pipelines**.
- Containerized application services using **Docker** and deployed on **AWS EC2**.

RECONCILE | *Python, FastAPI, PostgreSQL, Google OR-Tools, Groq (Llama 3.3)* *GitHub*

- Built a multi-tenant procurement platform (**68 REST endpoints**) where LLMs strictly draft text and summarize insights, while **deterministic engines** compute all pricing and landed costs.
- Modeled supplier allocation as a **MILP in Google OR-Tools** (capacity, MOQ, risk-weighted cost), solving 20 orders across 50 suppliers in **under 300 ms**; added a risk graph for disruption impact.

Flightdeck | *Java, Spring Boot, OpenTelemetry, Apache Kafka, Redis, PostgreSQL, Groq (Llama 3.1)* *GitHub*

- Engineered an observability platform ingesting **W3C-propagated OpenTelemetry spans via Kafka** to autonomously infer live service dependency topologies and reconstruct trace execution trees.
- Implemented online anomaly detection using **Welford's algorithm (**$O(1)$ **space)** and a **45-second sliding-window aggregator** to cluster cascading microservice failures into unified incident timelines.
- Formulated a **deterministic 6-factor scoring engine** isolating root causes from symptoms at **90%+ confidence**, paired with an automated replay harness achieving **100% sandbox failure reproduction**.

Experience

QA Team Lead — Havells Voice AI Beta Program **Feb–Jun 2026**

- Led a **4-member QA team** testing a voice-controlled AI agent for Havells smart appliances across BLE, Android, and iOS; reported **34 defects**, including **7 validated and 2 critical**.
- Collaborated with Havells engineers in weekly defect-review sessions within a 12-team QA effort that logged **532 defects**.

Certifications

ServiceNow Certified System Administrator (CSA) | ServiceNow Certified Application Developer (CAD)

Hackathons

I-SOLVE National Level Hackathon: Ranked among the **Top 10 teams** out of **746 participants**. Also participated in DSU DevHack 2.0 and the 24-hour Hackman v8 hackathon at DSCE.